

## Commercial TnH+IAQ Sensor with Modbus TCP/WIFI-MQTT

### Communication – Datasheet

#### 1. Product Overview

**WIN-SN-TnH-IAQ** is a versatile data acquisition device designed for real-time monitoring in industrial applications. Providing flexibility for various measurements. The device integrates seamlessly with **Modbus TCP**, offering reliable communication and long-range data transfer over several meters, making it ideal for large-scale installations.

Its robust design makes it suitable for **installations in HVAC systems, environmental monitoring, and process control**. With **configurable data sampling rates, multi-sensor compatibility, and easy configuration**, WIN-SN-TnH-IAQ ensures **accurate Air-quality, Temperature and Humidity data acquisition and reliable performance** in demanding industrial environments, making it an excellent choice for **industrial and IOT applications**. Additionally, it features a 4-digit 7-segment display that provides real-time measurement data, allowing users to monitor sensor readings directly without requiring an external interface.



Ex - Product Image

#### 2. Technical Parameters

- Measuring Parameters of Air Quality, Temperature, and Humidity.
- Response Time: Less than 2 seconds
- **Modbus TCP** communication

|                          | Sensor Name   | Temperature Measuring Range | Humidity Measuring range | AQI/ GAS/ Pressure              | Measuring Accuracy                    |
|--------------------------|---------------|-----------------------------|--------------------------|---------------------------------|---------------------------------------|
| <input type="checkbox"/> | <b>BME280</b> | -40 ~ 85°C                  | 0~100% RH.               | 300 - 1100 hPa                  | 1.0°C, 3.0 %RH,<br>1.0 %hPa           |
| <input type="checkbox"/> | <b>BME680</b> | -40 ~ 85°C                  | 0~100% RH.               | 300 - 1100 hPa<br>& 0 – 500 IAQ | 1.0°C, 3.0 %RH,<br>0.6 %hPa, 15 %IAQ  |
| <input type="checkbox"/> | <b>BME688</b> | -40 ~ 85°C                  | 0~100% RH.               | 300 - 1100 hPa<br>& 0 – 500 IAQ | 0.5°C, 3.0 %RH,<br>0.6 %hPa, 15 %IAQ  |
| <input type="checkbox"/> | <b>BME690</b> | -40 ~ 85°C                  | 0~100% RH.               | 300 - 1100 hPa<br>& 0 – 500 IAQ | 0.5°C, 3.0 %RH,<br>0.5 %hPa., 15 %IAQ |

|                          | Product Configuration |
|--------------------------|-----------------------|
| <input type="checkbox"/> | TnH+IAQ+E             |
| <input type="checkbox"/> | TnH+IAQ+E+D           |
| <input type="checkbox"/> | TnH+IAQ+MTCP+WiFi     |
| <input type="checkbox"/> | TnH+IAQ+MTCP+WiFi+D   |

### 3. Sensor Specifications

- **Wide temperature range:** -40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85\*85\*36 mm.
- **Certifications:**



## 4. Power Supply Connection

### DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

## Recommended Cable Electrical Characteristics

|                                          |                                           |
|------------------------------------------|-------------------------------------------|
| <b>22 AWG Cable</b>                      | Shielded and twisted pair should be used. |
| <b>Tinned Copper</b>                     | Recommended                               |
| <b>Nominal Conductor DCR</b>             | 14.7 ohm / 1000 ft                        |
| <b>Nominal Capacitance</b>               | 11 pf / feet (conductor to conductor)     |
| <b>High Frequency Non-Insertion Loss</b> | 0.5db / 100ft                             |

## 5. Step-by-Step Configuration

### 1. Connect the Device

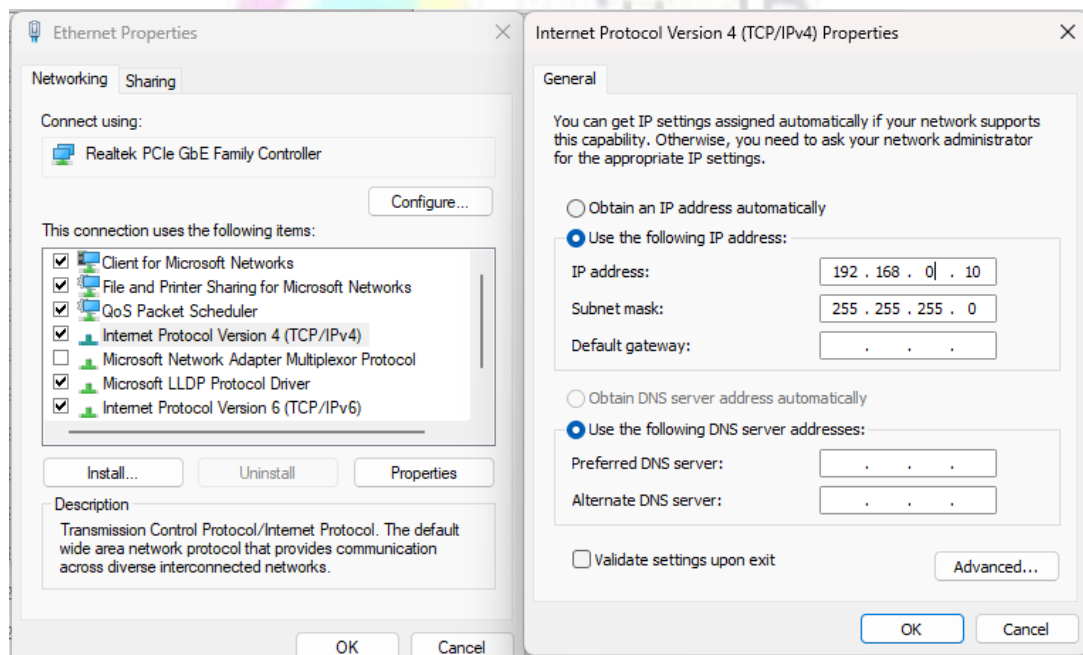


Image: 1

- Connect 12-24 VDC-1A DC adaptor to power the board
- Make the IPV4 configuration as 192.168.0.xx
- Enter 192.168.0.160 in web interface

## 1. Mode Selection

**WittelB-SN-Sensor**  
Configure your communication setting

Default IP: 192.168.0.160

Device Mode  


TCP-Mode  
WiFi-Mode

Subnet mask

Gateway

[Save Config](#)

**WittelB-SN-Sensor**  
Configure your communication setting

Default IP: 192.168.0.160

Device Mode

New IP Address

Subnet mask

Gateway

[Save Config](#)

- Enter **192.168.0.160** in the browser search bar and select the **Device Mode**.
- Select the device mode as **TCP-Mode** to get the sensor data via **Modbus TCP**, or select **WiFi Mode** to get the sensor data in the **MQTT cloud** through the local WiFi network.

### TCP Mode

- Here is an example of receiving data on Mb poll software.
- Open Mb-poll software connect using configuration setting (Default IP 192.168.0.160).
- After connecting you will get data as shown below.

**Connection Setup**

Connection  
☐ Serial Port ☒ TCP/IP ☐ UDP/IP

Port 9  
9600 Baud  
8 Data bits  
None Parity  
1 Stop Bit

Mode  
☒ RTU ☐ ASCII

Response Timeout  
500 [ms]

Delay Between Polls  
100 [ms]

Advanced...

Remote Server  
IP Address: 192.168.0.160 Port: 502 Connect Timeout: 3000 [ms]

[OK](#) [Cancel](#)

**Read/Write Definition**

Slave ID: 1

Function: 03 Read Holding Registers (4x)

Address mode  
☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 10

Scan Rate: 1000 [ms]

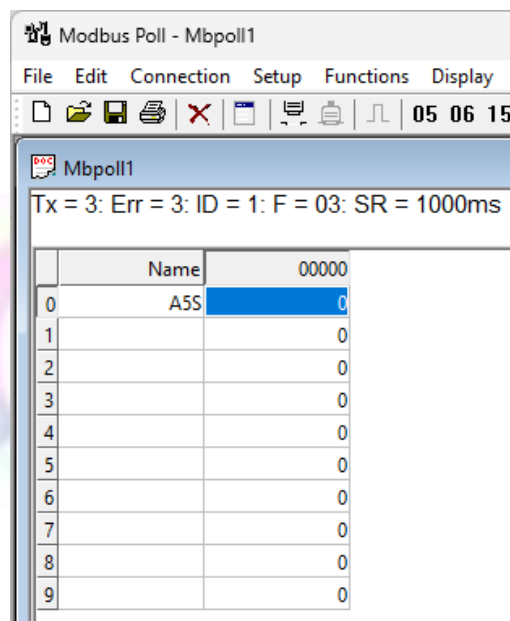
☐ Disable  
☐ Read/Write Disabled  
☐ Disable on error

View

[OK](#) [Cancel](#) [Apply](#) [Read/Write Once](#)

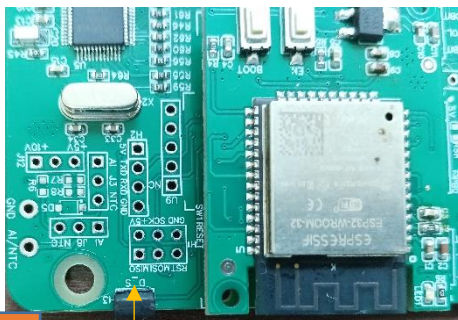
- Above image shows sensor data over Modbus TCP with default Modbus settings.
- Register 40001 contains the measured gas concentration value corresponding to the connected sensor (CO/CO<sub>2</sub>/NH<sub>3</sub>/PH<sub>3</sub>/H<sub>2</sub>S/CH<sub>4</sub>).
- Keep the jumper in the **single/open** as shown in image 1.2 position to change the IP address.
- Enter the desired IP address on the web page and click **Save**. The new IP address will be applied automatically.
- Connect using the new IP address to access the device and receive the data.

## Modbus Registers Mapping



|   | Name | 00000 |
|---|------|-------|
| 0 | A5S  | 0     |
| 1 |      | 0     |
| 2 |      | 0     |
| 3 |      | 0     |
| 4 |      | 0     |
| 5 |      | 0     |
| 6 |      | 0     |
| 7 |      | 0     |
| 8 |      | 0     |
| 9 |      | 0     |

| ID | Function Description | Register Description | Modbus Function Code | Protocol | Data Type  |
|----|----------------------|----------------------|----------------------|----------|------------|
| 1  | A5S Values           | 40001                | 0x03                 | TCP/IP   | 16-bit int |



**RESET  
JUMPER**

**Image: 1.1**



**IP config  
JUMPER**

**Image: 1.2**

## Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the above image 1.1) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

## WIFI Mode:

WittelB-SN-Sensor

Configure your communication setting

Default IP: 192.168.0.160

Device Mode

WiFi-Mode

WIFI SSID

WittelB

WIFI Password

augmatic@2020

Broker ID

broker.emqx.io

Publish Topic

tcp100

Username

admin

Password

public

Device ID

123

Port

1883

Publish Interval

10

Second

Second

Minute

**Image: 2**

+ New Subscription

tcp100

QoS 0

JSON

Topic: tcp100 QoS: 0

```
{
  "device": "11",
  "temp": "29.70",
  "hum": "26.59",
  "aqi": "93",
  "time": "15:49"
}
```

2026-03-28 17:36:45:532

Topic: tcp100 QoS: 0

```
{
  "device": "11",
  "temp": "29.78",
  "hum": "26.03",
  "aqi": "99",
  "time": "15:49"
}
```

2026-03-28 17:37:20:506

**Image: 3**

- Enter the local WiFi credentials as shown in above image2 so the device can connect to that WiFi network. Using this connection, it will send the sensor data to the cloud.
- Then enter the required MQTT Credentials and make sure the credentials should be correct.
- As shown in Image 2, you can select the publish interval in either minutes or seconds. MQTT data will be published according to the selected interval.
- Enter the MQTT credentials to transmit the sensor data to the cloud.
- Image 3 shows the sensor data at cloud for the selected topic.
- The published JSON payload includes a real-time timestamp, allowing each sensor reading to be tracked with its exact time.



### **Contact Information**

**Augmatic Technologies Pvt. Ltd.,  
Plot no 6, Shah Industrial Estate II,  
Kotambi,  
Vadodara – 391510.**

**Email – Sales@wittelb.com**

**Support: support@wittelb.raiseaticket.com / +91 94038 92805**